

Biosimilar Nivolumab Pricing Scenario

METIMMOX-1 Economic Evaluation

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1 Overview

This report presents cost-effectiveness results under a biosimilar nivolumab pricing scenario. The analysis examines whether reduced drug costs make biomarker-guided immunotherapy strategies cost-effective at the EUR 51,000 willingness-to-pay threshold.

1.1 Pricing Scenarios

We compare two pricing scenarios:

Table 1: Nivolumab Pricing Scenarios

Scenario	Nivolumab Cost	Price Change	Description
Base Case	EUR 13,923	–	List price for branded nivolumab per administration
Biosimilar	EUR 4,641	-67%	Biosimilar price (80% discount from list price)

The biosimilar price of EUR 4,641 per administration represents a 67% reduction from the base case price of EUR 13,923, equivalent to an 80% discount from the list price.

1.2 Model Structures

Each pricing scenario is analyzed using three alternative model structures for survival analysis:

Table 2: Model Structures for Survival Analysis

Model	Name	Formula Pattern	Description
Model A	Joint	Age + sex + Rx + crp*Rx + tlr*Rx + tmb_braf*Rx	All biomarkers and all treatment interactions
Model B	Focused	Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx	Per-strategy formulas: CRP/TMB_BRAF with both interactions
Model C	Separate	Age + sex + Rx + [biomarker]:Rx	One biomarker and its interaction per model

2 Deterministic Cost-Effectiveness Results

2.1 Multi-Model Comparison: Base Case vs. Biosimilar

2.1.1 Model A: Joint

Table 3: Model A: Joint - Base Case vs. Biosimilar Scenario

Strategy	Base Case			Biosimilar Scenario		
	Cost	QALY	ICER	Cost	QALY	ICER
Standard of Care	EUR 21,119	1.705	–	EUR 21,153	1.706	–
CRP-guided	EUR 54,941	1.813	EUR 182,853	EUR 32,057	1.813	EUR 65,796
TLR-guided	EUR 84,993	1.759	–	EUR 41,798	1.761	–
TMB/BRAF-guided	EUR 63,294	1.936	–	EUR 36,345	1.937	–

2.1.2 Model B: Focused

Table 4: Model B: Focused - Base Case vs. Biosimilar Scenario

Strategy	Base Case			Biosimilar Scenario		
	Cost	QALY	ICER	Cost	QALY	ICER
Standard of Care	EUR 21,155	1.703	–	EUR 21,123	1.703	–
CRP-guided	EUR 55,113	1.811	EUR 184,942	EUR 32,127	1.814	EUR 65,878
TLR-guided	EUR 85,187	1.753	–	EUR 41,836	1.756	–
TMB/BRAF-guided	EUR 63,639	1.932	–	EUR 36,300	1.934	–

2.1.3 Model C: Separate

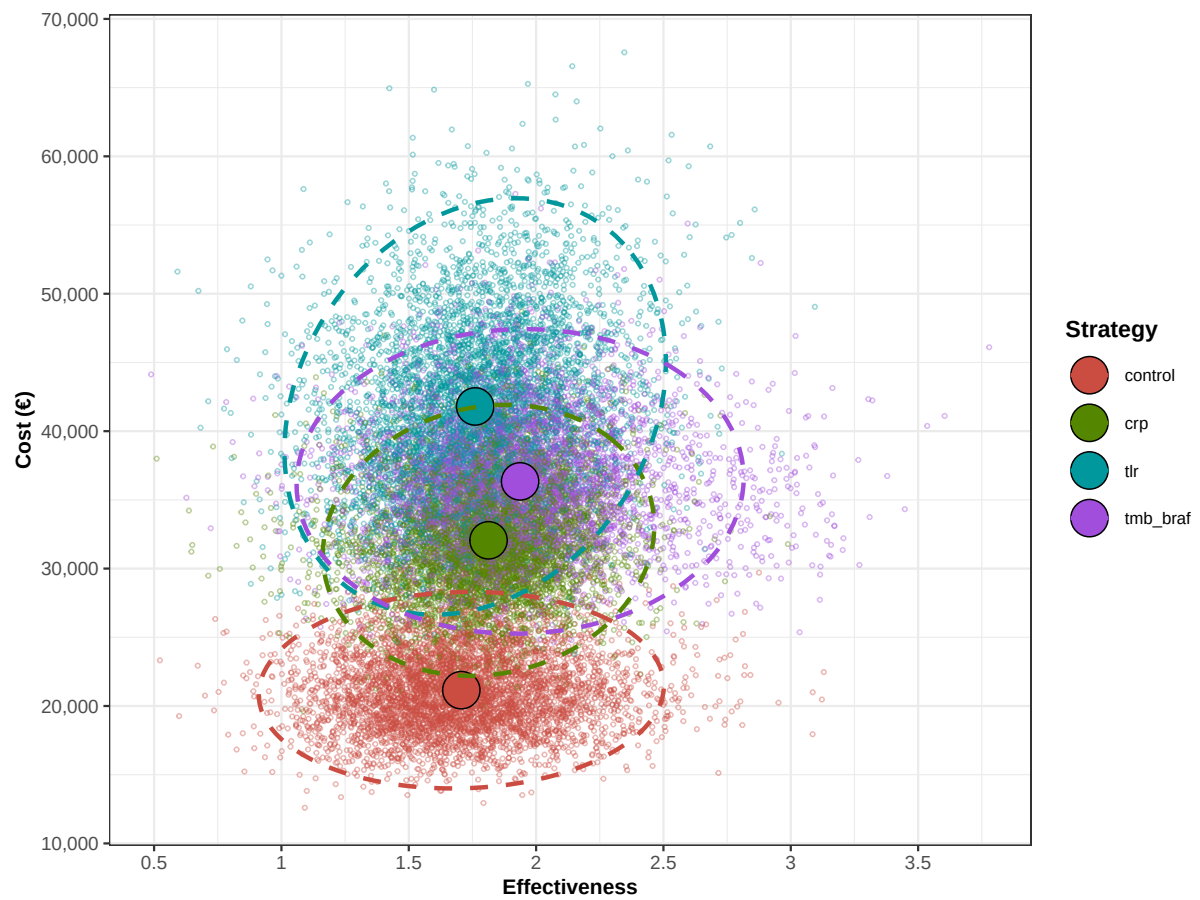
Table 5: Model C: Separate - Base Case vs. Biosimilar Scenario

Strategy	Base Case			Biosimilar Scenario		
	Cost	QALY	ICER	Cost	QALY	ICER
Standard of Care	EUR 21,125	1.705	–	EUR 21,134	1.705	–
CRP-guided	EUR 54,923	1.813	EUR 182,781	EUR 32,118	1.813	EUR 65,753
TLR-guided	EUR 84,928	1.757	–	EUR 41,778	1.758	–
TMB/BRAF-guided	EUR 63,377	1.936	–	EUR 36,264	1.935	–

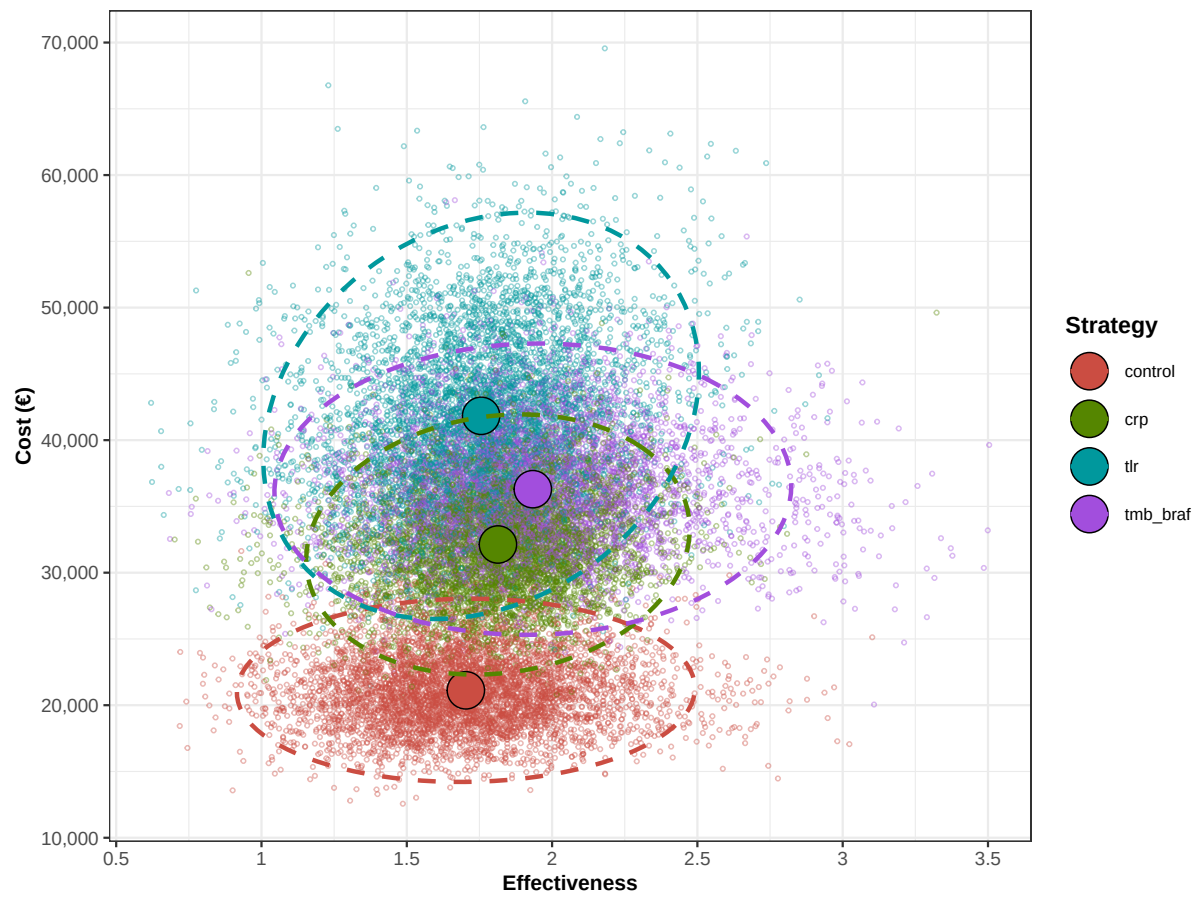
3 Probabilistic Sensitivity Analysis

3.1 Cost-Effectiveness Planes: Biosimilar Scenario

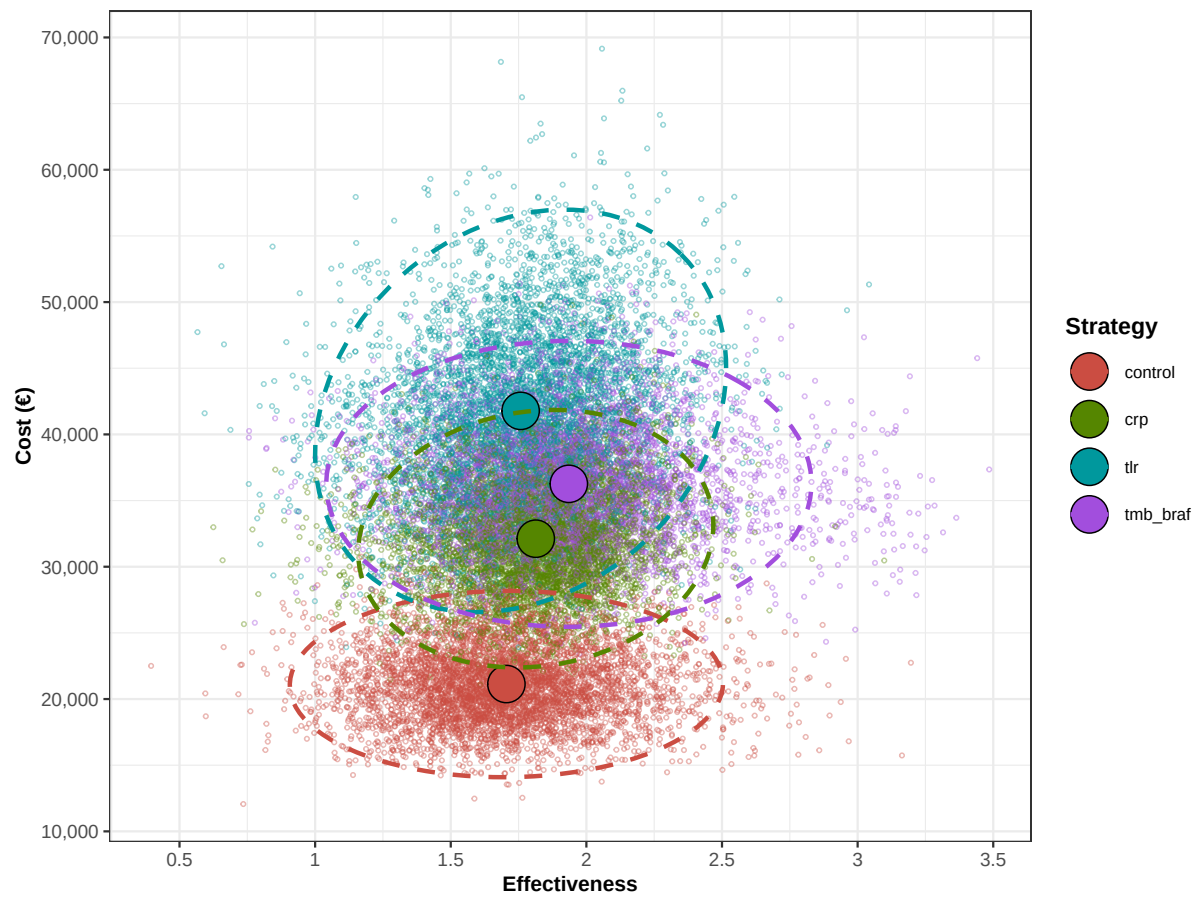
3.1.1 Model A: Joint



3.1.2 Model B: Focused



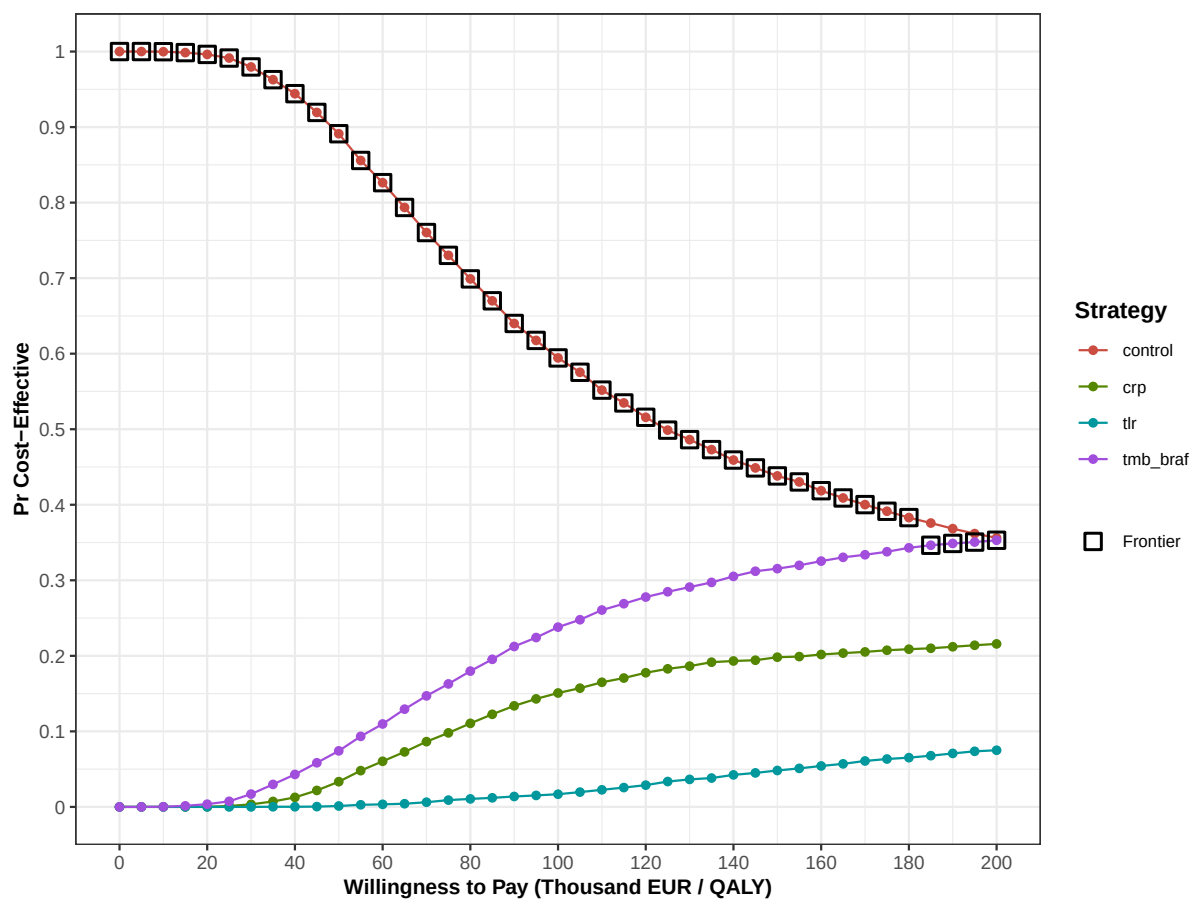
3.1.3 Model C: Separate



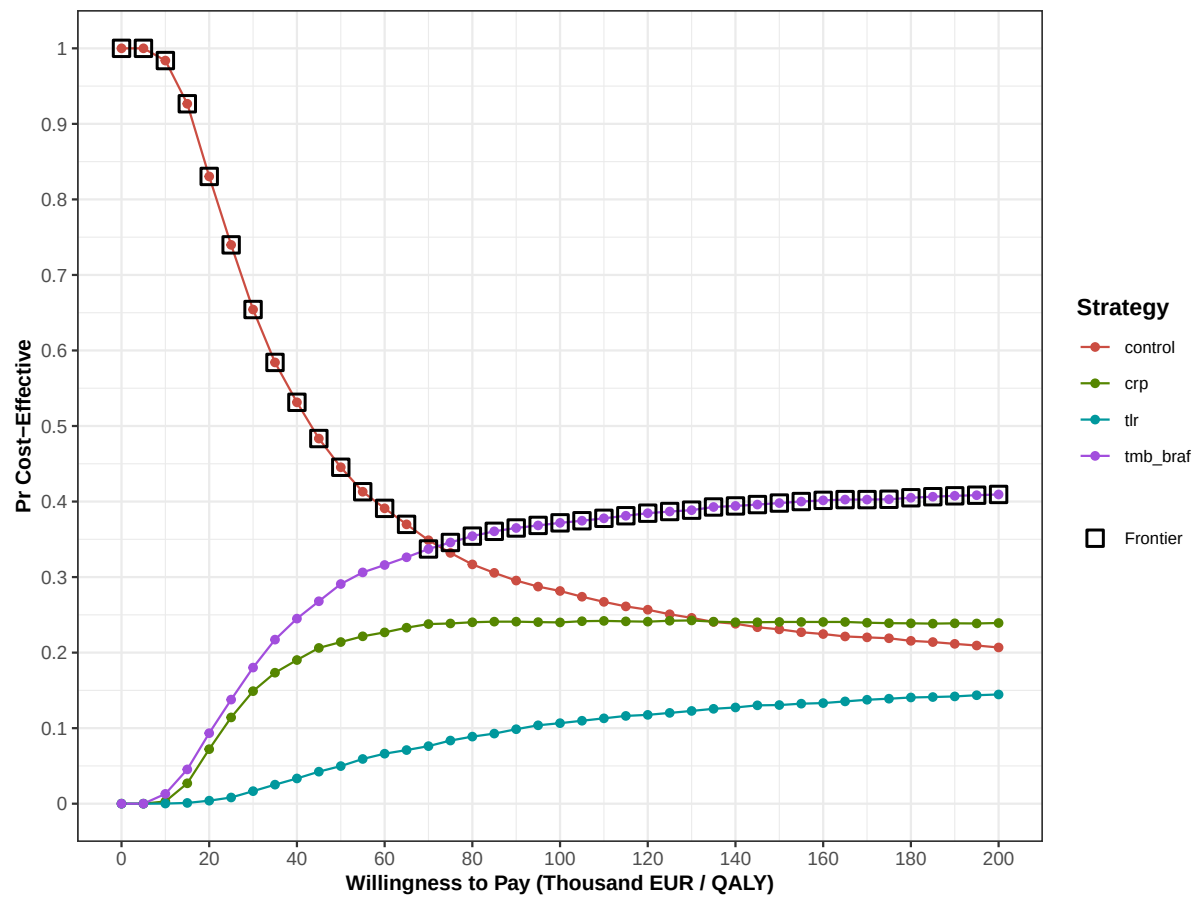
3.2 Cost-Effectiveness Acceptability Curves (CEAC)

3.2.1 Model A: Joint

Base Case:

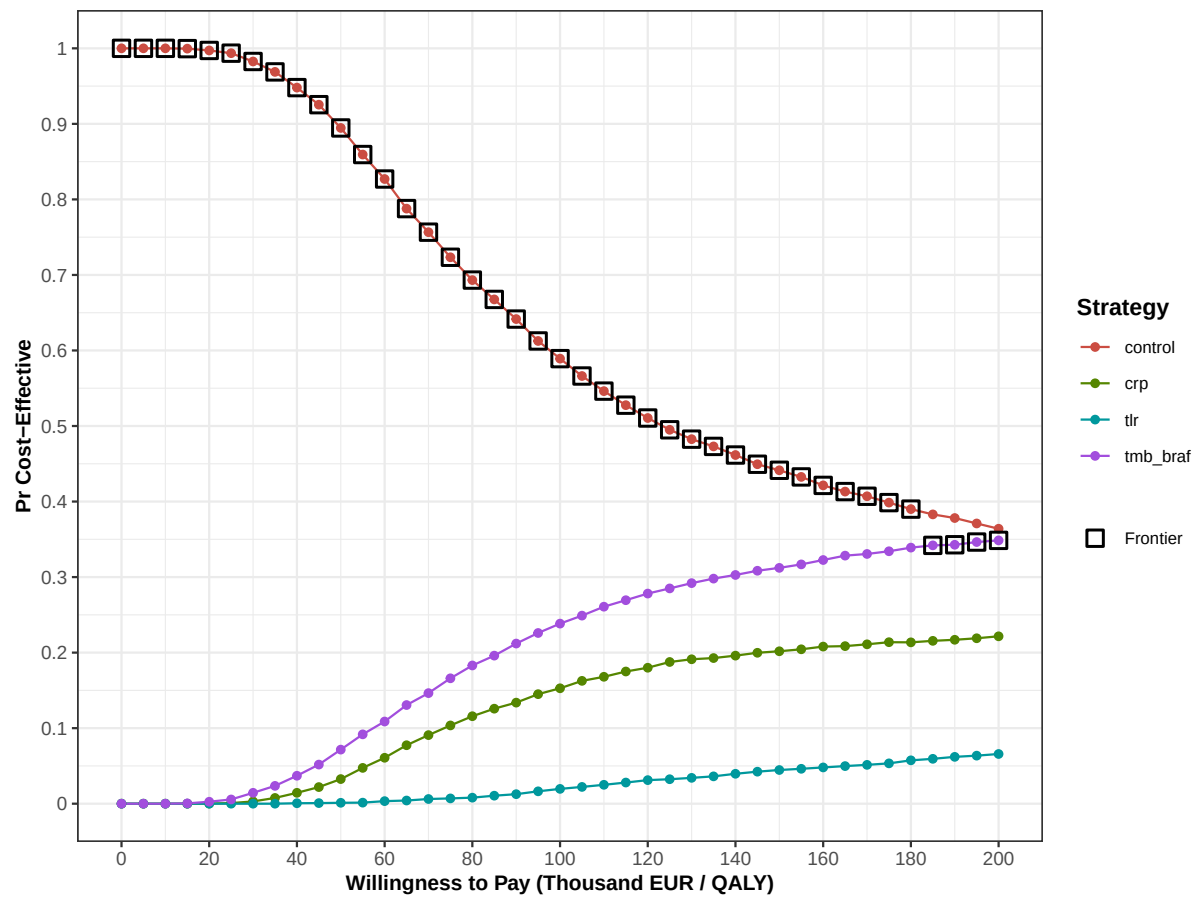


Biosimilar Scenario:

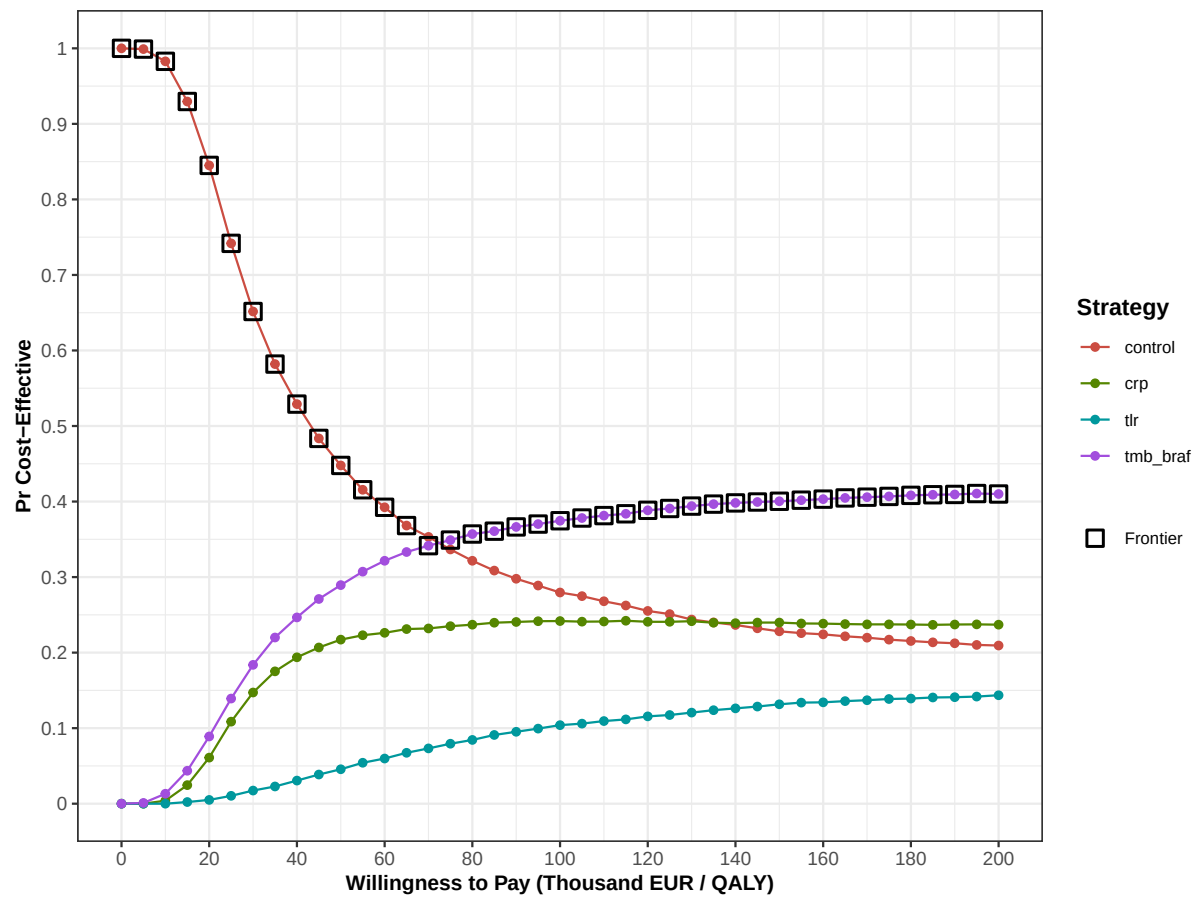


3.2.2 Model B: Focused

Base Case:

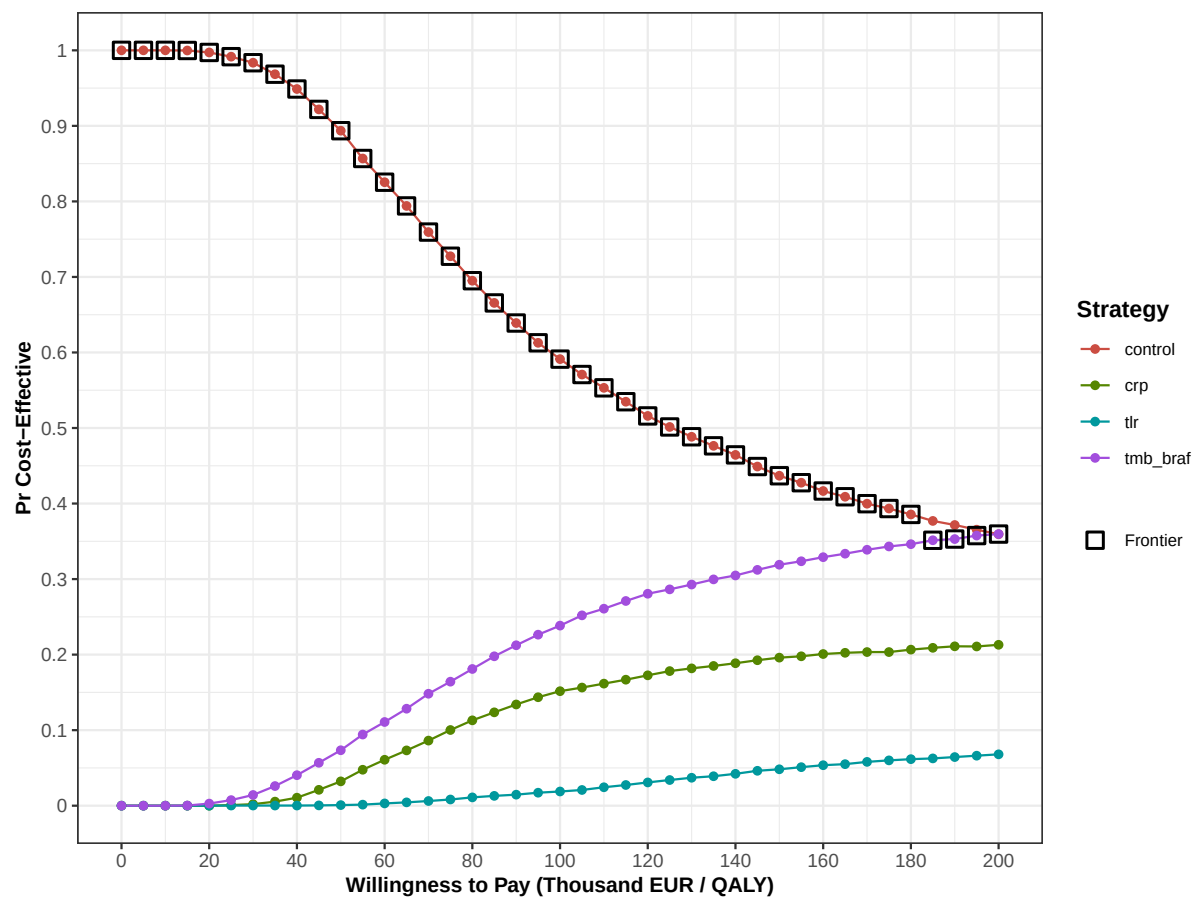


Biosimilar Scenario:

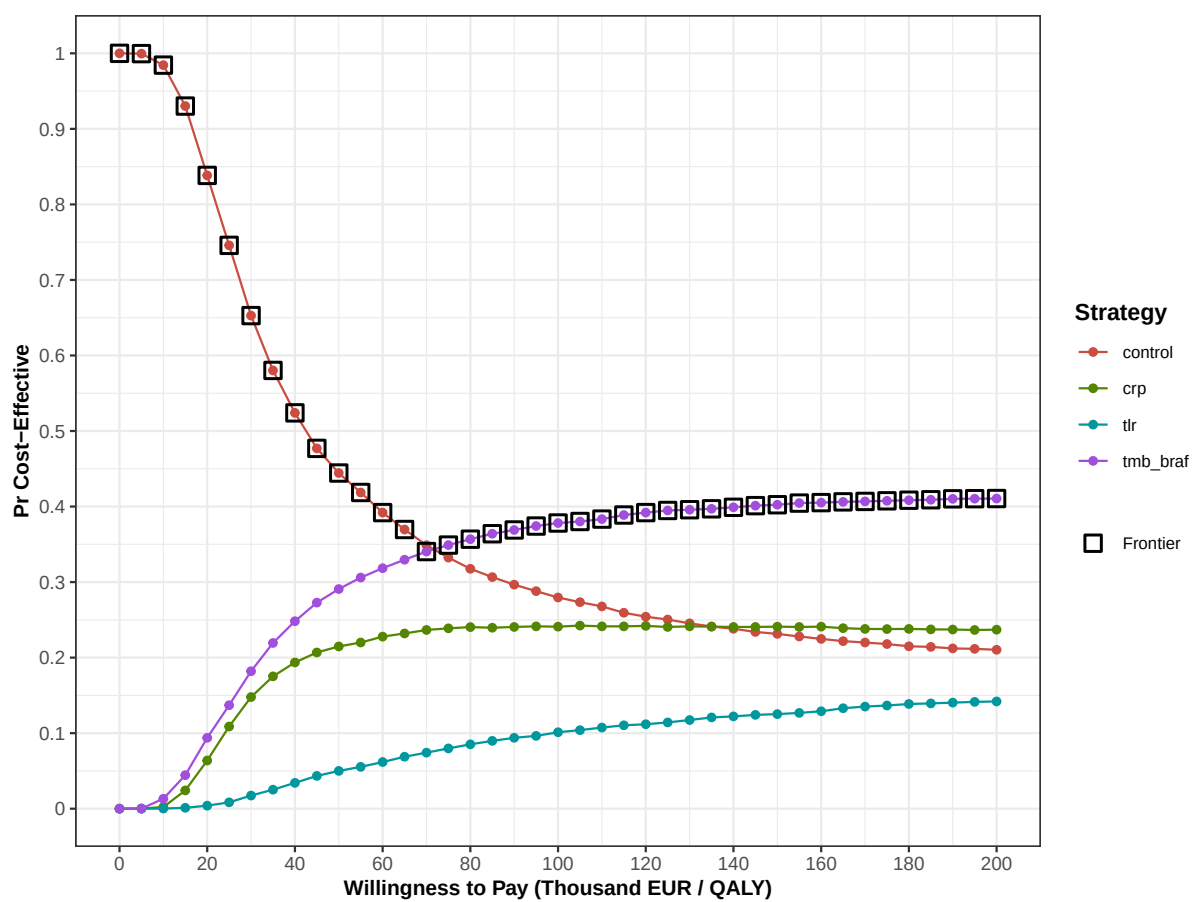


3.2.3 Model C: Separate

Base Case:



Biosimilar Scenario:



4 Summary of Key Findings

4.1 Cost-Effectiveness at EUR 51,000 WTP Threshold

Table 6: Probability of Cost-Effectiveness at EUR 51,000 WTP Threshold

Model	Strategy	Probability Cost-Effective		
		Base Case	Biosimilar	Change
Model A: Joint	Standard of Care	100.0%	98.4%	-1.6 pp
	CRP-guided	0.0%	0.3%	+0.3 pp
	TLR-guided	0.0%	0.0%	+0.0 pp
	TMB/BRAF-guided	0.0%	1.3%	+1.3 pp
Model B: Focused	Standard of Care	100.0%	98.3%	-1.7 pp
	CRP-guided	0.0%	0.4%	+0.4 pp
	TLR-guided	0.0%	0.0%	+0.0 pp
	TMB/BRAF-guided	0.0%	1.3%	+1.3 pp
Model C: Separate	Standard of Care	100.0%	98.4%	-1.6 pp
	CRP-guided	0.0%	0.2%	+0.2 pp
	TLR-guided	0.0%	0.0%	+0.0 pp
	TMB/BRAF-guided	0.0%	1.3%	+1.3 pp

4.2 Key Observations

Model A: Joint:

- Base case optimal strategy: Standard of Care (100.0% probability)
- Biosimilar optimal strategy: Standard of Care (98.4% probability)

Model B: Focused:

- Base case optimal strategy: Standard of Care (100.0% probability)
- Biosimilar optimal strategy: Standard of Care (98.3% probability)

Model C: Separate:

- Base case optimal strategy: Standard of Care (100.0% probability)
- Biosimilar optimal strategy: Standard of Care (98.4% probability)

4.3 Impact of Biosimilar Pricing

The 67% reduction in nivolumab cost (from EUR 13,923 to EUR 4,641 per administration) represents a substantial price decrease. However, the impact on cost-effectiveness depends on several factors:

1. **Direct cost savings:** Only biomarker-positive patients receive nivolumab, so cost savings are proportional to biomarker prevalence
2. **Testing costs:** Biomarker strategies incur additional testing costs (EUR 2,000 for TMB/BRAF NGS testing)
3. **Clinical effectiveness:** The biosimilar pricing scenario does not change clinical outcomes (QALYs), so any improvement in cost-effectiveness is purely through cost reduction

The results demonstrate whether the biosimilar price reduction is sufficient to make biomarker-guided strategies cost-effective at the EUR 51,000 willingness-to-pay threshold.

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